

Using ANI to Predict Molecular Energies

Introduction

Machine learning has become fundamental in all areas of science, including computational chemistry since it can make reliable approximations about expensive quantum mechanical calculations at much lower computational cost. As explained in the ANI-1 paper, accurate electronic structure methods are often too costly for large molecular systems, whereas faster approximate methods may result in a decline in accuracy.¹ Thus, a good approximation method is often a subtle balance between computational cost and accuracy. Nevertheless, neural networks can learn molecular energies from quantum mechanical reference data that predicts the molecular energies more efficiently after training.¹ This method is often proposed as a solution to the issue between computational cost and accuracy.

In the ANI-1 paper, an extensible neural network method (ANI) that uses atomic environment vectors (AEVs) was introduced by Smith, Isayev, and Roitberg to be trained on DFT energies for organic molecules containing H, C, N, and O atoms. By using AEVs to represent the local chemical environment, the ANI method was able to transform the molecular geometry into symmetry-based features that are invariant to translation, rotation, and permutation of identical atoms.¹ Without AEVs, raw Cartesian coordinates will be fed into the neural network. The transformation process is critical because the model should be transferable, which means that the total energy of a molecule should stay the same while the molecule is moved or rotated.

To predict the total molecular energy, the ANI model uses separate atomic neural networks for each atom type. It produces an atomic energy contribution as each atom's AEV passes through the neural network. It then calculates the total molecular energy by summing the predicted energy contributions over all atoms in the molecule, which makes the model more naturally scale to larger molecules. This method makes the model more transferable and efficient than standard neural networks that use flattened raw Cartesian coordinates or other fixed-size vectors.¹

In this project, TorchANI² was used in a supervised learning ANN model to predict molecular energies from the ANI dataset. The goal was to gain hands-on experience on building and running a training pipeline and to learn the workflow in training a ML model on real-life problems by tuning the model hyperparameters and architecture, testing the stability of the selected model, and finally training the selected model on all data provided in the project. The final result was then compared with the relevant RMSE values reported in the ANI-1 paper to make reasonable evaluations on the performance of the model.

Methods

In the ANI-1 paper, the reported full ANI-1 dataset is about 17.2 million conformations from about 58,000 molecules, with an 80/10/10 split for training, validation, and testing, respectively.¹ However, due to computational limits, the dataset used in this project was only a subset of the full ANI dataset.

The original dataset was obtained from molecules in the GDB database and contains organic molecules with C, H, O, and N atoms.¹ To predict molecular energy, the subset used in this project was first loaded using TorchANI², then self atomic energies were subtracted from the molecular energies. After the chemical species were converted into numerical values in the order H, C, N, and O, the dataset was shuffled.¹ In Checkpoint 2 of the project, the data were split with a proportion of 80/10/10 for training, validation, and testing, respectively, which allows the assessment of model performance on unseen data before tuning the final model.

Before passing through ANI, each atom was transformed into an atomic environment vector using TorchANI's AEV algorithm. The AEV length obtained from the AEV algorithm was 384. ANI used four separate atomic networks for H, C, N, and O, respectively.¹ During training, each atom's AEV were fed into the atomic neural network as the input features, and each atomic network had one hidden layer with hidden dimensions, which is followed by a ReLU activation and a final output layer.¹ The final output from each neural network represents the atomic contribution to the total molecular energy.

Since the project is a regression problem where the model predicts molecular energy values, the atomic neural network training was performed with mean squared error (MSE) loss and Adam optimizer.³ The model was mainly evaluated using mean absolute error (MAE) in kcal/mol. The final training on full data was evaluated using both MAE and root mean square error (RMSE) because relevant values in the ANI-1 paper were reported in RMSE, so RMSE was also computed for a fair comparison.

In Checkpoint 3, the project performed a tuning task to find the best model for the final full data training. The major hyperparameters tested were hidden dimension, learning rate, and L2 regularization, while batch size was fixed to 256. The hidden dimensions were tested with the values 64, 128, and 256. The learning rates were tested with the values 1e-3 and 1e-4. The L2 regularizations were tested with the values 0 and 1e-5. Thus, there were 12 combinations tested. For the comparison between hyperparameters to be fair, the model was reinitialized for each combination before training. This was essential because if the combinations were not reinitialized, later hyperparameter settings would be tested on a model already trained with previous hyperparameter settings.

The best configuration selected after hyperparameter tuning was the one with the lowest validation MAE (≈ 0.987 kcal/mol):

- hidden dimension = 64
- learning rate = 1e-3
- L2 regularization = 0

In Checkpoint 4, this best model was then used to test stability by performing repeated training runs and N-fold cross-validation. Lastly, this model was trained on full data in Checkpoint 5 for 20 epochs, and the final MAE and RMSE were reported as full-data fit metrics, not as held-out test metrics.

Results and Discussion

As shown in Table 1 below, the hyperparameter tuning results from Checkpoint 3, indicated that the model that performed best on the validation set was the one with hidden dimension 64, learning rate 1e-3, and no L2 regularization, which achieved a validation MAE of about 0.987 kcal/mol, the lowest among all other models. Table 1 also showed that validation performance was not improved by increasing the hidden dimension to 128 or 256. The reason might be that larger models usually need more computation and so will not always generalize well on a smaller dataset. In addition, the results showed that the learning rate of 1e-3 performed better than 1e-4. During training, models with 1e-3 also converged faster than the ones using 1e-4. One more information that Table 1 gives was that L2 regularization did not improve performance and generalization for this model and dataset.

Table 1. Hyperparameter tuning results from Checkpoint 3.

	hidden_dim	learning_rate	l2	final_val_loss	train_mae	val_mae
0	64	0.0010	0.00000	0.000005	0.979937	0.987342
1	256	0.0010	0.00000	0.000005	0.983503	0.994686
2	128	0.0010	0.00000	0.000006	1.146211	1.152172
3	256	0.0001	0.00000	0.000011	1.189181	1.201160
4	128	0.0001	0.00000	0.000010	1.326049	1.346883
5	64	0.0001	0.00001	0.000013	1.425070	1.433479
6	256	0.0001	0.00001	0.000013	1.431747	1.442144
7	64	0.0001	0.00000	0.000014	1.515486	1.535375
8	128	0.0010	0.00001	0.000015	1.570841	1.579133
9	128	0.0001	0.00001	0.000017	1.688814	1.693933
10	256	0.0010	0.00001	0.000019	1.720675	1.723294
11	64	0.0010	0.00001	0.000019	1.720143	1.728336

The model selected was then retrained using a 80/10/10 split for train/validation/test and was evaluated on the held-out test set with test MAE around 1.10 kcal/mol. Figure 1 below showed the predicted vs. true energy plot. As shown in Figure 1, most points were close to the diagonal $y = x$ line, which means that the model predictions were close to the reference energies. This indicates that the ANI-style atomic neural network model was able to learn the relationship between molecular structure and reference energy for the course dataset and that the selected model predicts molecular energies with low error.

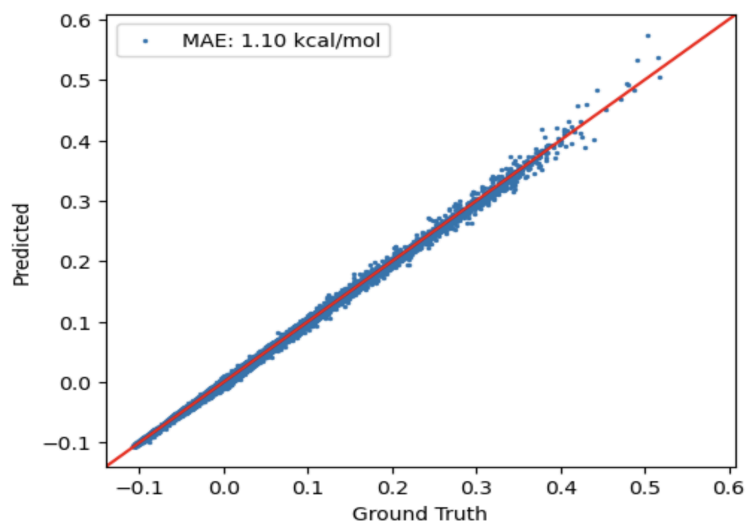


Figure 1. Predicted energy vs. true energy for the selected model from Checkpoint 3. Most points lie close to the $y = x$ line, showing that the selected model predicts molecular energies with low error.

In Checkpoint 4, the stability of best configuration obtained from Checkpoint 3 was tested using multiple runs and 5-fold cross-validation. In the multiple-run experiment, 5 runs were conducted with the same train/validation/test split of 80/10/10 splitting. As shown in Table 2 below, different runs produced slightly different MAE values. This indicates that random initialization and training randomness affect the final model, which is expected since neural networks depend on stochastic optimization. Nevertheless, the average test MAE remained below the 2 kcal/mol test MAE provided in the final project guideline, which indicates that the chosen model is working well.

In the N-fold cross-validation, 5 folds were conducted where the data were split into 5 different train/validation folds. Cross-validation was conducted to confirm that the model was generalized instead of remembering features from any specific train/validation split. As shown in Table 2, the mean training and mean validation errors are almost the same. These results show that the model was reasonably stable and was able to generalize well. Since the purpose of CV is to rotate the validation set across developmental data, the CV algorithm only evaluated training and validation folds, so the statistics for mean test MAE were NaN.

Table 2. Checkpoint 4 multiple-run and cross-validation summary.

	Experiment	mean_Train_MAE	std_Train_MAE	mean_Val_MAE	std_Val_MAE	mean_Test_MAE	std_Test_MAE
0	Multiple Runs	1.754949	0.390384	1.751838	0.390835	1.759738	0.387445
1	5-Fold CV	1.434535	0.353518	1.438651	0.351070	NaN	NaN

In Checkpoint 5, the selected model from Checkpoint 3 was trained on all available data. Since the model was trained on full data, there was no separate held-out test set for this final model. Instead, the final full-data MAE and RMSE were calculated as fit metrics.

As shown in Figure 2 below, the training loss decreased over the 20 final training epochs, indicating that the final model was improving consistently during full-data training.

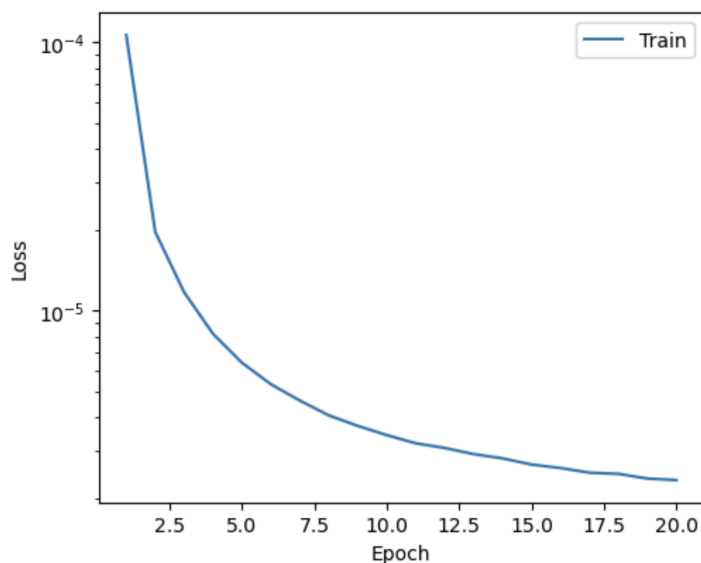


Figure 2. Checkpoint 5 training loss curve. The loss decreases over training.

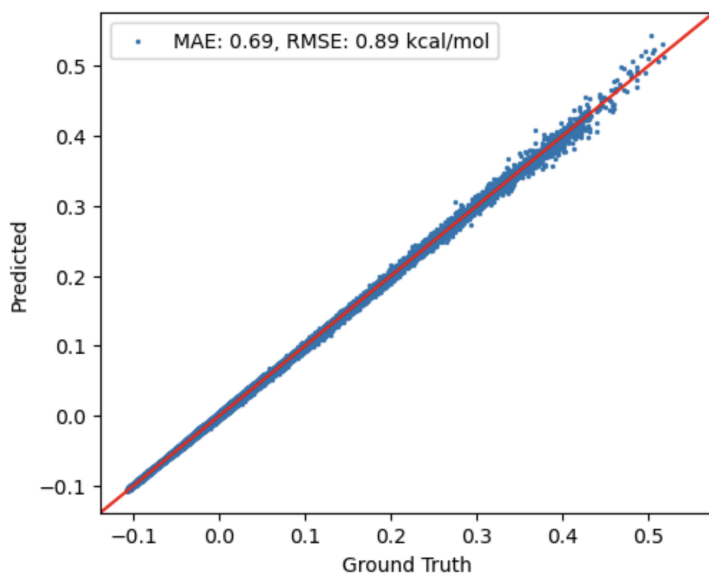


Figure 3. Final Checkpoint 5 full-data prediction plot. The final model was trained on all available data, and the predicted energies were compared with the reference energies. The model achieved a full-data MAE of 0.694 kcal/mol and a full-data RMSE of 0.893 kcal/mol.

As shown in Figure 3 above, the final predicted vs. true energy plot showed that most points lie close to the diagonal $y = x$ line, and the final full-data MAE obtained was 0.694 kcal/mol with a final full-data RMSE of 0.893 kcal/mol, both are relatively low errors. This indicates that the final model fits the project dataset well after training on all available data.

The ANI-1 paper reported the model-fit RMSE values as 1.2 kcal/mol for training, 1.3 kcal/mol for validation, and 1.3 kcal/mol for testing. In comparison to these values, the final full-data RMSE of 0.893 kcal/mol is lower. However, this is not a one-to-one comparison because the full-data RMSE was evaluated on one full dataset while the model-fit RMSE reported by ANI-1 paper was split-based values on a larger and more diverse dataset of about 17.2 million conformations from about 58,000 molecules. The ANI-1 model was also more complex and had a much larger data-generation pipeline. Thus, the lower full-data RMSE does not represent a better model in comparison to the ANI-1 model. It only indicates that the selected model works well on the course dataset, and can be seen as a successful implementation of the ANI idea.

Overall, the validation, test and CV results from Checkpoint 4, combined with the full-data training result in Checkpoint 5, supported that the selected model learned the dataset patterns sufficient enough that the model was able to predict molecular energies with relatively low error. This supports the effectiveness of AEV-based atomic neural networks for molecular energy prediction.

Conclusion

In conclusion, this project successfully reproduced the main ANI idea that molecular energies can be predicted by converting local atomic environments into AEVs, passing them through atom-specific neural networks, and summing atomic energy contributions. In Checkpoint 3, hyperparameter tuning selected the best configuration of hidden dimension 64, learning rate $1e-3$, and no L2 regularization with the lowest validation MAE among all other configurations. The selected model was then retrained using a 80/10/10 split and evaluated on the held-out test set with test MAE around 1.10 kcal/mol. The selected model also showed reasonable stability in multiple-run and cross-validation experiments from Checkpoint 4. Lastly, the selected model was trained on all available data and achieved a full-data MAE of 0.694 kcal/mol and a full-data RMSE of 0.893 kcal/mol. Even though this result is not appropriate for direct comparison to the model-fit RMSE values reported in the ANI-1 paper, it still Overall, the results show that the ANI framework can successfully learn molecular energy patterns from AEV features and produce predictions on the course dataset with high accuracy.

References

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3. Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. arXiv:1412.6980.